# Igor Augusto de Melo

Aeronautical Engineering student
São Carlos School of Engineering (EESC-USP) · São Carlos, SP, Brazil

igor_amelo@usp.br
linkedin.com/in/igorademelo
github.com/igor-de-melo
igor-de-melo.github.io/aero/

## Summary

**Aeronautical Engineering** student at EESC-USP, with an exchange year at École Centrale de Marseille, currently pursuing a special studies certificate in **CFD** at ICMC-USP. Experience in propulsion, flight mechanics, aerostructures and systems engineering, with **three titles at the Latin American Space Challenge (LASC)**, and project manager of an autonomous fixed-wing UAV at the SEMEAR group. Research in LES, WRF and unsteady aerodynamic loads; interested in flight control, embedded systems and cybersecurity.

## Education

- **Undergraduate degree in Aeronautical Engineering**, *EESC, University of São Paulo* — 2020 – expected 2027
- **Academic exchange, general engineering program**, *École Centrale de Marseille, France* — Aug 2022 – Sep 2023
- **Special Studies Certificate in Computational Fluid Dynamics**, *ICMC-USP* — 2023 – present

## Experience

**Project Manager, autonomous fixed-wing UAV** — Apr 2025 – present
*SEMEAR Aerial Robotics Group, EESC-USP*

- Manage an **autonomous fixed-wing UAV** project in **preliminary design**: a 14-person team across 5 subsystems.
- Define the architecture and **technical requirements** and lead validation using **NASA systems engineering**.
- Work on aerodynamics and flight dynamics: aerodynamic analysis, performance and control-oriented simulations.

**Data Analyst Intern** — Dec 2024 – Mar 2025
*Pilea Labs, influencer and affiliate marketing platform for e-commerce*

- Designed and built the **MySQL** databases on **Google Cloud** and the queries behind the platform's charts.
- Joined sales by influencer coupon with social media metrics, at a scale of **thousands of orders**.
- Proposed about 15 dashboard KPIs, delivered **real-time** analyses and integrated **Python** scripts into the website.

**Team Member: propulsion, flight mechanics and systems engineering** — Mar 2020 – Dec 2023
*TOPUS Projetos Aeroespaciais, EESC-USP*

- Designed, tested and analyzed **solid and hybrid** rocket motors as a team, using thermodynamics, heat transfer, structural analysis and **CFD**.
- Built 6-DOF flight simulation tools in **MATLAB** and **Monte Carlo** dispersion analysis.
- Applied **NASA** systems engineering methods to internal project management and documentation standards.

**Structures Team Member** — Mar 2017 – Jan 2018
*Keep Flying, AeroDesign team (Poli-USP)*

- Sized the structure and computed the loads of a maximum-**payload** aircraft, with weight/payload trade-off studies.

## Aerospace Projects

**Competition rockets at LASC** — 2020 – 2022
*TOPUS, Latin American Space Challenge*

- Won, with the team, **three titles in three editions**: Curupira (2020, 1st place in the Entry-Level 0.5 km category, 8th overall) and Rigel (1 km, 2021 and 2022).
- Rigel 2022 motor (team work): K-class solid motor with **KNSB** propellant; calculated values of 2,418 N·s total impulse and 127.8 s specific impulse.
- Simulated the Rigel 2022 flight (MATLAB, 6 DOF): simulated apogee of 1,160 m; **apogee reached: 1,168 m**.
- Ran **5,000 Monte Carlo simulations** (thrust, drag, wind) to map the landing area; simulated mean apogee of 1,153 m.

**Boitatá hybrid rocket motor** — 2020 – 2023
*TOPUS, oxidizer injection team*

- Designed and tested the tank, piping and valves, and designed the $N_2O$ **vortex injector** (60 bar tank, 30 bar chamber).
- Ran **CFD** of the injector in **ANSYS CFX**; project awarded a Brazilian Space Agency (AEB) test stand grant in 2020.

**Rocket with a 5 km apogee** 2020 – 2022

*LASC Cooperation Program, aerostructures team*

- Produced the nose cone drawings and manufacturing plans, supported **composite** studies and ran **2D/3D CFD** (**ANSYS Fluent**).

## Research

**Turbulent flow simulation** Sep 2023 – Dec 2024

*ICMC-USP, FAPESP grant*

- Simulated turbulent channel flow with Coriolis force using **LES** ($Re_\tau$ = 180, 64 × 64 × 64 grid), compared with **DNS**.
- Developed and ran the numerical routines in **Fortran** on HPC; results published at EPTT 2024.

**Unsteady aerodynamic loads** Sep 2023 – Dec 2024

*Undergraduate research, EESC-USP, advised by Prof. Flávio Donizeti Marques*

- Developed semi-empirical models of unsteady aerodynamic loads, calibrated against experimental and **CFD** data.
- Implemented a **neural network** without ML libraries in **Python/NumPy** and **MATLAB** to optimize the model parameters.

**Atmospheric modeling and wind energy research** Feb – Jul 2023

*IRPHE, École Centrale de Marseille, France*

- Set up and ran the **WRF** weather model on three nested domains (Python, Bash, Linux).
- Analyzed wind-related uncertainty in a wind turbine's energy output (defense slides, in French).

## Publication

Igor Augusto de Melo, Livia S. Freire, Leandro Franco de Souza, Matheus Tozo de Araujo. *Turbulent Channel flow with Coriolis force using Large-Eddy Simulation*. 14th Spring School on Transition and Turbulence (EPTT 2024), São João da Boa Vista, SP, Brazil, Sep 2024. Paper (PDF).

## Data and Automation

- **Email triage** (personal project): **Claude Haiku** through the Claude API classifies emails into JSON, with prompt caching; Python, OAuth2, IMAP, cron and Telegram on a VPS.

## Skills

- **Engineering:** aerodynamics, CFD, propulsion, flight mechanics, aerostructures, control systems (fundamentals).
- **Simulation:** ANSYS Fluent/CFX, OpenFOAM, XFOIL, XFLR5, WRF; FEA: ANSYS Mechanical, Patran/Nastran and Abaqus (basic).
- **Programming:** Python, MATLAB/Simulink, Fortran, SQL, Bash, C++ (basic), Java (basic); Git, Linux, HPC.
- **Tools:** SolidWorks, Fusion 360, CATIA (basic), Google Cloud, LaTeX.
- **Management:** systems engineering (NASA SE Handbook), requirements, technical documentation, traceability.

## Courses and Certificates

- Machine Learning and Reinforcement Learning courses, Data ICMC-USP (2025); Finite Element Analysis with MATLAB; XFOIL online short course; GOM Inspect Suite basic training.
- Project Management MOOC (Gestion de Projet), Centrale Lille: System Operation Diagnosis and Advanced Planning (Nov 2022); LASC participation certificates (2020, 2021 and 2022).

## Languages and Other Activities

- **Languages:** Portuguese (native), English (advanced), French (advanced), German (basic), Spanish (basic).
- **Volunteering:** Gatos do C2, USP São Carlos (Mar – Oct 2024); coordinated rescue logistics for the spay/neuter, care and release program.